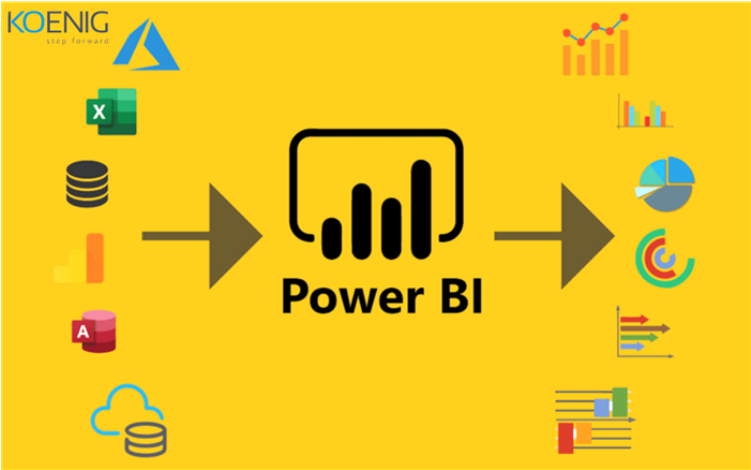
POWER BI AND SQL INTEGRATION

What is POWER BI?

'Power Business Intelligence' constitutes the full form of Power BI, a Business Intelligence tool crafted by Microsoft. It allows users to visualize and share insights from their data. For Excel users, Power BI's user interface is pretty intuitive, and its strong connectivity with other Microsoft applications makes it a flexible self-service tool with minimal initial training needed. With Microsoft power bi you can quickly connect to data prep models and visualize as much or as little as you like. Then securely share insights even embed it all in your app or website to get started download the free power bi desktop application and how to write in soon you'll be connecting hundreds of the data sources on premise or in the cloud like Excel, Salesforce, Google analytics, social networks and even IoT devices to get real time information.



Turning all that data into live interactive visuals. When you get the hang of that upgrade to pro, you'll need to start collaborating freely, rating dashboards that deliver a 360-degree real time view of your business, control how your data is accessed to use and package and deliver your stunning new content to users. With apps of their own the premium version is for really scaling up across larger enterprises and embedding content past your own digital properties without per user licenses and one more thing you want it all on mobiles so you can react to changes on the fly. There you go with power bi we make it easy to always be complying too for things like GDPR. Download the free power bi application and start turning your data into smarter decisions.

The applications power BI at different areas,

- Business analytics
- Data visualization
- Measure ROI
- Optimize marketing campaigns

What is SQL?

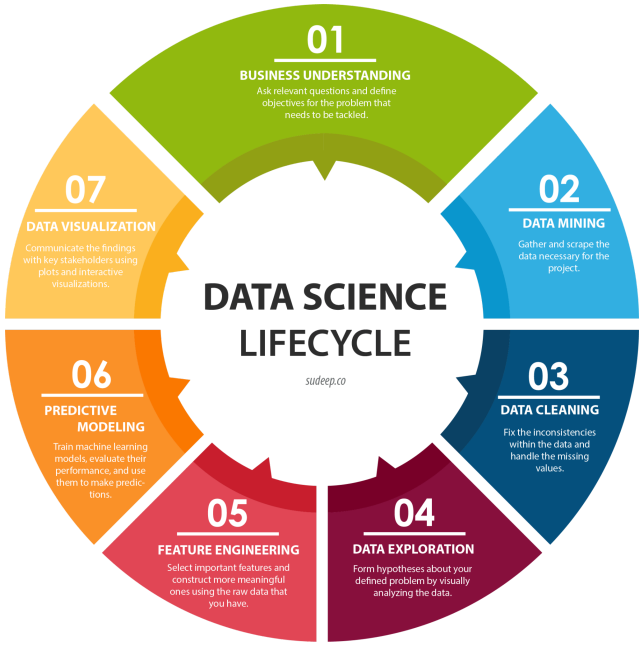
SQL is a well known programming language used for storing and processing information in relational databases. SQL stands for 'Structured Query Language.' In relational databases the information is stored in the form of tables which consist of rows and columns. This will represent different data attributes and various relationships between the data values. Its syntax is easy to read, so it's easy to pick up on even if you're completely new to programming, and it's even useful for non-technical careers.

Data analysts and developers learn and use SQL because it integrates well with different programming languages. For example, they can embed SQL queries with the Java programming language to build high-performing data processing applications with major SQL database systems such as Oracle or MS SQL Server. SQL is also fairly easy to learn as it uses common English keywords in its statements.

SQL is widely used in software development, data analysis, business intelligence, and data warehousing. It's essential for tasks such as data retrieval, manipulation, reporting, and database administration. Using SQL as the primary API for relational databases by big data platforms and organizations is standard.

What is Data Science?

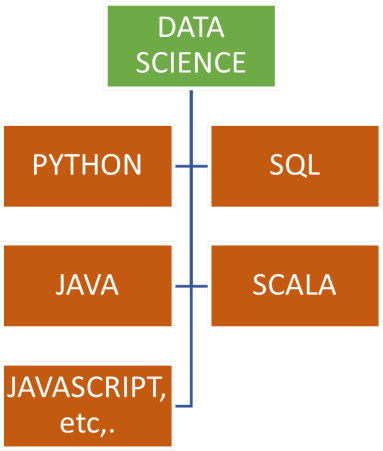
The field of data science studies how to use large amounts of data using contemporary tools and methods to uncover patterns, extract valuable information, and make business decisions. Data science creates predictive models by utilizing sophisticated machine learning techniques. The information utilized for analysis can be found in a variety of formats and originate from a wide range of sources. After learning what data science is, let's examine the data science way of life.



This is a visualization of the life cycle for Data Science. Data science is becoming one of the most popular fields in the technology business quite quickly. We can now analyze large datasets thanks to the speedy improvements in computer efficiency, which have made it possible to find previously unheard-of patterns and insights regarding user behavior and global trends.

How is SQL related to DATA SCIENCE ?

SQL is a programming language which comes under DATA SCIENCE technology. We all know that the data scientists need to access and retrieve the data for their operations. For this purpose SQL helps the data scientists to do their work easily. Data science involves extracting value and insights from large volumes of data to drive business decisions. It also involves building predictive models using historical data.



Elements of SQL for Data Science:

Following are the key points of SQL that are most helpful for Data Science. Every Data Scientist must know these necessary SQL skills and features.

- Relational Database Model
- SQL Query Commands
- Handling Null Values
- Joins
- Key Constraints
- Working with SubQuery
- Creating Tables and Databases

We must extract data from the database in order to work with it. SQL enters the picture in this situation. A key component of data science is relational database management. A data scientist can define, define, create, and query the database using SQL commands. Many different industries and organizations have used NoSQL to manage their product data, yet SQL is still the best choice for many.

What is the importance of SQL in data science?

SQL has a good significance in Data Science due to its Scalability, Interoperability, Performance optimization, and Large data exploration. SQL is a good fundamental skill for data scientists since it enables efficient data retrieval, cleaning, exploration, integration, and analysis. Mastering SQL empowers data scientists to work with diverse datasets and extract valuable insights to drive data-driven decision-making processes. It helps data scientists access, manage, and analyze large data sets to derive insights that can drive better business decisions. Here are some of the things which we can do on integration.

Data Extraction

Data scientists use SQL to extract specific data from a database for analysis. Moreover, they use SQL queries to filter and select data based on different criteria such as time, location, or other variables.

Data Transformation

SQL cleans and transforms data, creating new tables or views that are easier to analyze. It can be used to merge, join, or concatenate tables, as well as calculate new variables.

Data Aggregation

Using SQL, you can calculate summary statistics, group data into specific categories such as age and gender, and calculate averages like net national income and gross domestic income.

Data Visualization

SQL can be used to create tables and views that are easier to visualize using data visualization tools like Tableau, Power BI, or other data visualization software.

Data Exploration

In terms of data exploration, SQL for data science professionals is important to identify patterns and relationships between different variables. With SQL queries, you can filter data and calculate different metrics that help with further exploratory data analysis.

POWER BI and SQL Integration?

Most of the data scientists integrate SQL with POWER BI desktop for the data visualization. This allows the user to access real-time data directly from your database, enabling up-to-date insights and analysis. Additionally the SQL server's advanced capabilities integrate machine learning and predictive analysis.

We can integrate the POWER BI and SQL in many ways,

Import Data: Here POWER BI takes the data from SQL server and stores the data locally in the form of POWER BI file. After this we can visualize the data in our required form. This method is useful when you want to work with a snapshot of the data or when real-time access is not required.

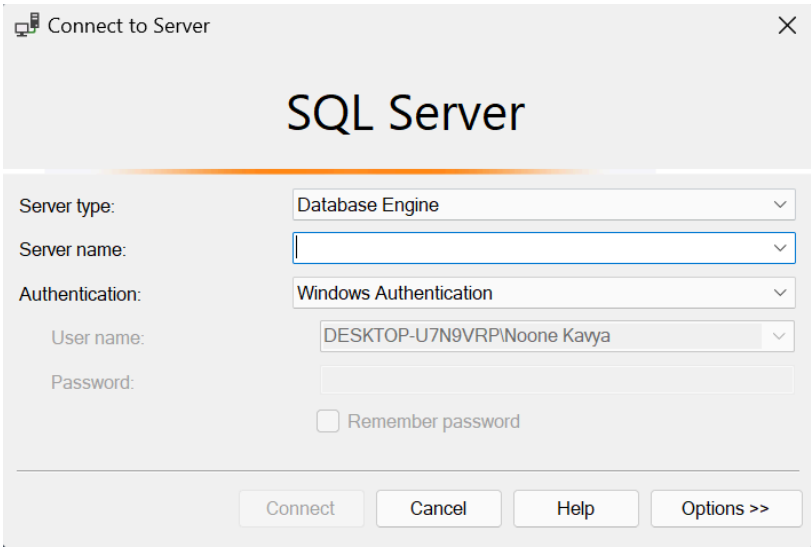
Direct Query: Power BI sends queries directly to the SQL Server database in real-time, retrieving the most up-to-date data for visualization and analysis.

Data Modeling: Power BI enables you to create relationships between tables imported from SQL Server and perform data modeling tasks within Power BI Desktop. You can define relationships, create calculated columns and measures, and enrich your data model to support complex analytics and visualizations.

There are a lot of ways to integrate POWER BI and SQL server. These are top used ways of integration. Here I have performed a real time data visualization which will be explained step by step with the help of snapshots and the important queries used in the SQL server for the database creation along the output from the POWER BI.

Detailed explanation of POWER BI and SQL server integration:

Step 1: First you need to have a database for this visualization. If you have you can use your own database, if you don't have a database follow these steps to create a database on your own. Install SQL server management system on your device, and if you are a new user, you will get a pop up window to connect the SQL server. Enter your server name to connect the SQL server.



Step 2:

Click the 'New Query' to get a new script file to write the queries to create a database and perform actions to it.

```sql
CREATE DATABASE HRDB;
GO

USE HRDB;
GO
```

These queries will create DATABASE named 'HRDB' and this database will be used for further modifications. Now we need to write the data in this HRDB database.

```sql
CREATE TABLE EMPLOYEES
(
ID INT IDENTITY PRIMARY KEY,
First_Name varchar(20),
Last_Name varchar(30) not null,
email varchar(30) unique,
salary int
);
GO
```

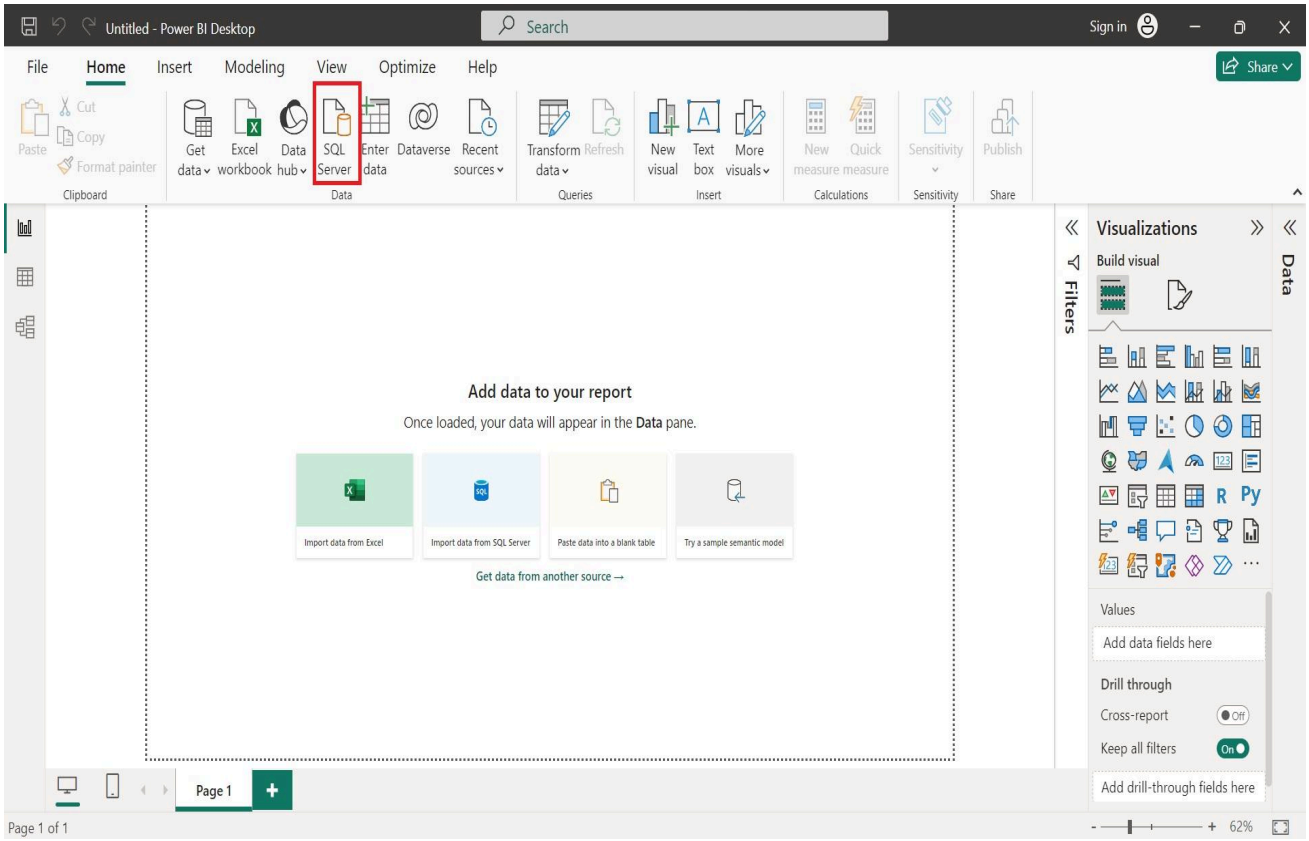
These queries will initiate the variables which are going to be used in this database. And now we need to provide these variables with the required data for the visualization.

```sql
INSERT INTO EMPLOYEES(First_Name, Last_Name, email, salary)
VALUES
  ('John', 'Doe', 'john.doe@example.com', 50000),
  ('Jane', 'Smith', 'jane.smith@example.com', 60000),
  ('Michael', 'Johnson', 'michael.johnson@example.com', 30000),
  ('Emily', 'Brown', 'emily.brown@example.com', 52000),
  ('James', 'Davis', 'james.davis@example.com', 58000),
  ('Sarah', 'Wilson', 'sarah.wilson@example.com', 40000),
  ('Matthew', 'Martinez', 'matthew.martinez@example.com', 45000),
  ('Emma', 'Anderson', 'emma.anderson@example.com', 57000),
  ('David', 'Taylor', 'david.taylor@example.com', 35000),
  ('Olivia', 'Thomas', 'olivia.thomas@example.com', 55000);
```

Here is the random information which is written for the data visualization for integration between POWER BI and SQL server.

With these steps we have created our own database and now we need to integrate this database with POWER BI.

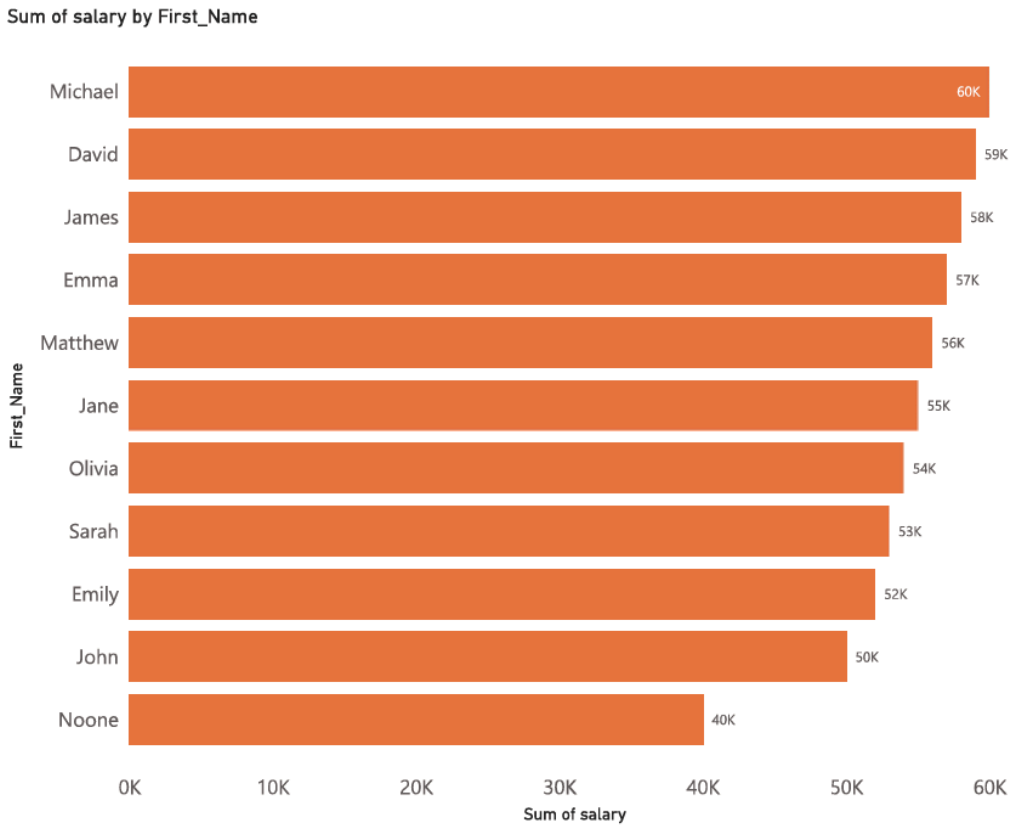
Step 3:



This is POWER BI view and to connect with the Sql click the boxed SQL server and enter the server name which you want to connect with the database. After connecting you will be able to view the databases which are created in that server.

You can modify the data by selecting the required number of rows which can be termed as data cleaning. In POWER BI we can visualize the imported data in our required format like Graph, Bar Graph, Pie Chart and so on.The POWER BI helps the analyst to do their job easily with an interactive model.

Step 4: Now you can perform your required visualization on the data to get models.



This is the visualization for my case. Which gives me the information about employees salary mentioned with their first name in the increasing order. You can visualize according to your requirements.

Conclusion:

SQL is one of the well known programming languages used for the DATA SCIENCE and the analysts are integrating the SQL scripts with the POWER BI for the interactive visualization model.SQL is a good fundamental skill for data scientists since it enables efficient data retrieval, cleaning, exploration, integration, and analysis. Mastering SQL empowers data scientists to work with diverse datasets and extract valuable insights to drive data-driven decision-making processes.

It helps data scientists access, manage, and analyze large data sets to derive insights that can drive better business decisions. Here in this blog the process of integrating the SQL and POWER BI is clearly explained step by step along with the snapshots of the sql queries and snapshots.

References:

simplilearn.com/tutorials/data-science-tutorial
sudeep.co/data-science
aws.amazon.com/what-is/sql
process.st/how-to/connect-power-bi-to-sql-server
geeksforgeeks.org/sql-for-data-science/
emeritus.org/blog/data-science-and-analytics-sql-for-data-science/
How to Create Database in SQL Server